Evaluation of StreetLight AADT Estimates of Iowa Interstates

Accuracy and Independence Assessment

Contents

1 Executive Summary

This report evaluates StreetLight Data's AADT estimates for Iowa interstate segments. Its focus is twofold: evaluate the methodological independence from Iowa DOT counts, and assess estimation accuracy relative to FHWA and Minnesota DOT benchmarks. The findings are intended to inform how Iowa DOT weighs the trade-offs between reduced tube-count deployment and the accuracy limitations StreetLight exhibits.

On independence: Available documentation supports StreetLight estimates being primarily probe-driven, with Iowa DOT counts used for calibration and validation rather than direct segment-level estimation. This cannot be confirmed without explicit vendor disclosure, however, and the more consequential concern may be operational: Reducing physical tube counts would eliminate the independent baseline needed to detect future (potential) degradation in StreetLight accuracy over time.

On accuracy: Performance varies substantially by volume range.

AADT Range	MAPE (2022 / 2024)	Relative to FHWA Guidance
$\geq$ 20,000	8.6% / 6.8%	Within bounds
10,000–19,999	11.0% / 9.6%	At or near upper limit
7,500–9,999	13.0% / 15.3%	Exceeds bounds
$<$ 7,500	47.5% / 38.7%	Far exceeds bounds

StreetLight estimates are consistently higher than Iowa DOT counts across all volume ranges — a directional tendency that is modest at higher volumes but pronounced at lower ones.

Segment-level diagnostics identify 7 persistently problematic segments (APE $>$ 100% in both years) and a pool of 8 eligible segments with exceptionally close agreement (APE $\leq$ 1% both years). The latter represents a natural starting point for a potential monitoring strategy that would preserve validation capability at reduced cost. Full details are provided in Section 5.

2 Study Purpose and Research Questions

This report evaluates the performance of StreetLight Data's Average Annual Daily Traffic (AADT) estimates for Iowa interstate segments by comparing them to observed Iowa DOT counts. Two primary questions are examined:

1. **Methodological Independence:** To what extent are StreetLight estimates produced independently of Iowa DOT counts versus calibrated against them?
2. **Accuracy:** How closely do StreetLight AADT estimates match Iowa DOT counts across different traffic volume ranges?

The overall goal is not only to assess current agreement, but also to understand what kind of strategy would allow Iowa DOT to maintain confidence in Streetlight AADT estimates over time.

2.1 Question 1: Independence of StreetLight Estimates

Available evidence supports StreetLight estimates being primarily probe-driven, with Iowa DOT counts used for calibration and validation rather than direct segment-level estimation. Whether observed agreement fully reflects methodological independence — or is partly an artefact of Iowa DOT data having been used during model training — cannot be determined without vendor disclosure. The practical implications of this uncertainty, and the case for retaining a targeted set of physical counts regardless of how the independence question is ultimately resolved, are explored in Section 3.

2.2 Question 2: Accuracy of Streetlight Estimates

StreetLight AADT estimates perform reliably on higher-volume Iowa interstate segments and are broadly consistent with independent validation benchmarks from Minnesota DOT. Specifically for road segments carrying 10,000 or more vehicles per day (the primary traffic volume group of interest), average errors range from roughly 7% to 11%, which falls within acceptable bounds under both MN DOT and FHWA guidance. Accuracy degrades at lower volume ranges, and segments below 7,500 AADT exhibit error levels too high to support direct substitution for official counts.

Across all volume buckets, StreetLight estimates tend to run slightly higher than Iowa DOT counts. This directional tendency is consistent and modest on higher-volume roads, but becomes more pronounced at lower volumes.

Overall, the results support using StreetLight AADT estimates as a credible, independently generated source for higher-volume Iowa interstate segments, with the recommendation that low-volume segments be treated with appropriate caution. These results point to an important practical question: How should DOT balance the cost savings of reduced counter deployment against the accuracy limitations StreetLight exhibits at lower volumes? The findings here suggest a one potential strategy — leveraging StreetLight for high-volume segments while maintaining physical counters where estimates are less reliable.

Note: MN DOT applies a data quality filter before computing its published accuracy metrics — a step not applied here — so some portion of the gap between Iowa and Minnesota results is methodological rather than reflective of true differences in estimate quality. This distinction is detailed in Section 4.2.

3 Question 1: Independence

3.1 Conceptual Framing

StreetLight estimates and DOT counts measure the same underlying quantity: true average annual traffic volume. Agreement between the two is therefore expected and, by itself, does not establish methodological dependence.

The relevant question is whether DOT counts are used *directly* to produce segment-level estimates, or whether they are used only for calibration and validation of probe-based models.

3.2 Summary Assessment

Current evidence supports StreetLight estimates being primarily probe-driven rather than directly derived from Iowa DOT counts. The concern that Iowa DOT is effectively "paying to receive its own data back" appears overstated based on available documentation. The more substantive and underappreciated risk is different: *Reducing or eliminating Iowa DOT tube counts would degrade both parties' ability to assess whether StreetLight estimates are accurate for Iowa segments in the future.*

This matters for three reasons:

- **DOT loses independent validation.** Without ground-truth counts on a representative set of segments, there is no external check on StreetLight accuracy. Observed agreement in this report is itself conditional on having DOT counts to compare against.
- **StreetLight loses Iowa-specific calibration signal.** DOT counts are used for model calibration and plausibility screening. Removing them degrades the model's ability to self-correct for Iowa-specific conditions.
- **The source of accuracy becomes unverifiable.** For any given segment, it is currently impossible to determine whether a close StreetLight estimate reflects genuine probe-data quality or is partly an artefact of Iowa DOT counts having been used during calibration. Without ongoing counts, this ambiguity is permanent and estimates must be taken on faith.

The question of methodological independence is therefore secondary to the practical question of *what monitoring practice preserves the ability to detect and respond to degraded accuracy over time.*

To summarize the state of evidence:

- **What can be concluded:** Available documentation supports StreetLight estimates being primarily probe-driven, with DOT counts used for calibration and validation rather than direct segment-level estimation.
- **What cannot be concluded:** Whether observed agreement reflects genuine methodological independence or is partly an artefact of Iowa DOT data having been used during model training — a distinction that requires vendor disclosure to resolve.
- **Operational implication:** Regardless of how the independence question is ultimately answered, retaining a targeted set of tube counts on well-chosen segments preserves the ability to detect and respond to degraded accuracy over time.

3.2.1 Implications

- Strong observed agreement between DOT counts and StreetLight estimates is expected and is not diagnostic of dependence, as both measure the same underlying quantity
- Statistical tests cannot resolve whether agreement reflects probe quality or calibration influence

- Definitive confirmation would require vendor disclosure that Iowa DOT data are not used as direct inputs to segment-level estimation, and/or Iowa-specific holdout validation
- One possible path forward is a targeted retention strategy for tube counts on a small set of well-chosen segments (see *Recommended Monitoring Segments*, Section 5.4 below)

3.3 Details

The following provides further detail on the basis for this assessment: why agreement is expected, how independence should be evaluated, what evidence is currently available, and what uncertainties remain.

3.3.1 Why Agreement Is Expected

When comparing StreetLight estimates to Iowa DOT observed counts, high correlation is both observed and expected because both aim to measure the same latent quantity: true average annual traffic volume.

- High correlation is therefore necessary but not diagnostic.
- The relevant question is not *whether* the estimates agree, but *why* they agree.

3.3.2 What Methodological Independence Means in Practice

Methodological independence depends on *how counts enter the modeling pipeline*, not whether they appear anywhere in it.

The practical question is:

> To what extent are Iowa DOT counts used to *produce* StreetLight Iowa estimates, rather than simply evaluate them?

Current evidence indicates that Iowa DOT counts are used for calibration and validation rather than direct estimation, though documentation does not fully rule out the use of Iowa-specific count data during model training.

4 Question 2: Differences in Estimates

4.1 Recommended Bucketing

MN DOT buckets are used here to allow for direct comparison against an independent benchmark. The four buckets and their AADT ranges are as follows:

Bucket Name	AADT Range
Sub-7,500	$< 7{,}500$
Lower-volume evaluation	7,500 – 9,999
Medium-volume evaluation	10,000 – 19,999
High-volume evaluation	$\geq 20{,}000$

The 2022 and 2024 conflated datasets are treated as **separate validation exercises** against the same underlying segment definitions, providing two measurement cycles against which consistency of results can be evaluated, while noting that the repeated use of the same segments across years is partly what motivates the misclassification analysis in Section 4.3.2. The road segments in those files correspond directly to Iowa interstates of interest; the original Iowa DOT data, while more numerous, included segments outside the scope of this work. Throughout this report, Iowa DOT counts serve as the reference standard for evaluation purposes, while recognizing that physical tube counts are themselves subject to measurement error.

4.2 Accuracy Assessment

Comparing StreetLight AADT estimates against Iowa DOT counts, accuracy is broadly consistent with MN DOT's 2024 validation in terms of error magnitude, though Iowa estimates are systematically higher across all buckets. As noted in Section 2.2, MN DOT applies a percent-change tolerance screen not used here — specifically, excluding locations where estimated AADT deviates more than 15% (volumes 5,000–49,999) or more than 10% (volumes $\geq 50{,}000$) — which accounts for some portion of this gap. Past work has historically focused on the Mean Absolute Percent Error (MAPE) metric for comparing the accuracy of estimates. This measure is used throughout in addition to a handful of other established measures of accuracy.

For the $\geq 20{,}000$ AADT bucket, MAPE ranges from 6.8% (2024) to 8.6% (2022), modestly above MN DOT's reported 5.81% but well within the FHWA Table 14 upper bound of approximately 10.5% for medium-volume roads (5,000–54,999) at the observed sample sizes. Signed percent error is positive at approximately 4.3% to 4.6%, in contrast to MN DOT's slight negative bias of -3.25%, indicating a mild tendency toward overestimation in the Iowa data; median TCE likewise falls within the FHWA $\pm 2.0\%$ bias limit at large n. NRMSE is 9.0% to 10.7%, and APE SD is 6.9% to 9.1%. Taken together, these error levels are low in absolute terms, consistent with reliable probe-based estimation on higher-volume roadways, and meet FHWA accuracy standards. *Among the volume buckets examined, this group provides the strongest basis for using StreetLight estimates in place of physical counts.*

In the 10,000 to 19,999 bucket, MAPE is 9.6% to 11.0%, compared to MN DOT's 6.28%. This range approaches but does not exceed the FHWA Table 14 upper bound of approximately 10.5–11.9% for medium-volume roads at the observed sample sizes — placing these results at the upper limit of what FHWA guidance would consider acceptable, with little margin remaining. Signed errors are positive (4.8% to 5.8%) versus MN DOT's near-zero 0.26%, again reflecting a tendency for overestimation, though median TCE remains within or near the FHWA ± 2–4% bias limits depending on sample size. APE SD (11.9% to 13.8%) is approximately double MN DOT's 4.98%, suggesting higher variability across segments even where average accuracy is marginal. *StreetLight estimates in this range may be suitable for planning purposes where moderate error is tolerable, but should not be treated as direct substitutes for official counts without additional validation.*

In the 7,500 to 9,999 bucket, MAPE is 13.0% to 15.3%, somewhat above MN DOT's 12.16% and exceeding the FHWA Table 14 upper bound of approximately 10.5–12.7% for medium-volume roads at the observed sample sizes. Signed errors are positive at 7.3% to 10.2%, a notable directional contrast to MN DOT's −4.02% and substantially above the FHWA ±2–5% bias limits. APE SD is elevated at 18.5% to 22.0% versus MN DOT's 8.30%, indicating considerable segment-to-segment variability. Together these figures suggest this bucket does not meet FHWA accuracy standards as stated. *Estimates in this range should be used with caution and interpreted alongside, rather than in place of, available count data.*

For the Sub-7,500 bucket — not reported in the MN DOT study — errors are substantially higher and merit particular caution. MAPE ranges from 38.7% (2024) to 47.5% (2022), and NRMSE from 49.1% to 66.3%. These figures far exceed the FHWA Table 14 upper bound of approximately 13–18% for low-volume roads (500–4,999) at the observed sample sizes, and the observed bias similarly exceeds the FHWA ±1.6–8.7% TCE median limits. Notably, the 2022 APE SD of 150.3% indicates extreme variability across segments in that year, well above the more moderate 46.6% observed in 2024. The magnitude of this difference likely reflects one or more high-leverage outlier segments in the 2022 sub-7,500 pool rather than a general year-over-year shift in model behavior; the flagged segments in Section 5.2 are a natural starting point for investigation. Regardless, error levels in this bucket fail FHWA accuracy standards by a wide margin, and StreetLight estimates below 7,500 AADT should not be substituted for official counts without substantial additional scrutiny. *This bucket falls outside the range where StreetLight estimates can be considered reliable for any planning or reporting purpose without independent verification.*

Note: More details on MN DOT's plausibility screen are described in the *2024 Mobile Source Data for AADT Implementation Analysis* (Methodology section).

Applying a similar filter to the Iowa data would likely reduce MAPE estimates across buckets, closing some — though not necessarily all — of the gap between Iowa and Minnesota results. That filter was not applied here for two reasons: first, it requires a prior-year official IADOT count for each segment against which to assess percent change, and complete prior-year counts were not available for the full set of conflated segments; second, presenting unfiltered results provides a more conservative and complete picture of StreetLight performance across the Iowa interstate network, avoiding any implicit imputation of which segments would have passed a tolerance screen.

4.3 Metrics Comparison

4.3.1 Summary

Table 3: Iowa AADT Accuracy Summary — 2022

AADT Bucket	n	MAPE (%)	Median % Error	Mean % Error	APE SD (%)	NRMSE (%)
Sub-7,500	394	47.47	13.45	36.11	150.29	66.25
(7,500-9,999)	93	12.96	4.21	7.34	18.46	22.76
(10,000-19,999)	157	10.98	6.87	5.75	13.76	16.31
(20,000+)	59	8.59	4.41	4.63	9.13	10.70

Table 4: Iowa AADT Accuracy Summary — 2024

AADT Bucket	n	MAPE (%)	Median % Error	Mean % Error	APE SD (%)	NRMSE (%)
Sub-7,500	400	38.71	23.99	32.12	46.60	49.14
(7,500-9,999)	92	15.29	8.01	10.18	21.96	25.63
(10,000-19,999)	156	9.57	4.95	4.79	11.94	14.62
(20,000+)	60	6.79	3.04	4.25	6.87	8.96

4.3.2 Bucketing

Table 5: AADT Bucket Comparison — 2022

Bucket	IADOT n	StreetLight n	Difference (STL - IADOT)	% Difference
Sub-7,500	394	377	-17	-4.3
(7,500-9,999)	93	86	-7	-7.5
(10,000-19,999)	157	175	18	11.5
(20,000+)	59	65	6	10.2

Table 6: AADT Bucket Comparison — 2024

Bucket	IADOT n	StreetLight n	Difference (STL - IADOT)	% Difference
Sub-7,500	400	369	-31	-7.8
(7,500-9,999)	92	104	12	13.0
(10,000-19,999)	156	173	17	10.9
(20,000+)	60	62	2	3.3

Table 7: AADT Bucket Misclassification — 2022

Category	Count	% of Total Obs	% of Misclassified
Stayed the Same	634	90.2	NA
Total Misclassified	**69**	**9.8**	**NA**
Sub-7,500 → 7,500-9,999	20	2.8	29.0
Sub-7,500 → 10,000-19,999	6	0.9	8.7
Sub-7,500 → 20,000+	0	0.0	0.0
7,500-9,999 → Sub-7,500	5	0.7	7.2
7,500-9,999 → 10,000-19,999	24	3.4	34.8
7,500-9,999 → 20,000+	0	0.0	0.0
10,000-19,999 → Sub-7,500	4	0.6	5.8
10,000-19,999 → 7,500-9,999	2	0.3	2.9
10,000-19,999 → 20,000+	7	1.0	10.1
20,000+ → Sub-7,500	0	0.0	0.0
20,000+ → 7,500-9,999	0	0.0	0.0
20,000+ → 10,000-19,999	1	0.1	1.4

Table 8: AADT Bucket Misclassification — 2024

Category	Count	% of Total Obs	% of Misclassified
Stayed the Same	635	89.7	NA
Total Misclassified	**73**	**10.3**	**NA**
Sub-7,500 → 7,500-9,999	30	4.2	41.1
Sub-7,500 → 10,000-19,999	5	0.7	6.8
Sub-7,500 → 20,000+	0	0.0	0.0
7,500-9,999 → Sub-7,500	2	0.3	2.7
7,500-9,999 → 10,000-19,999	21	3.0	28.8
7,500-9,999 → 20,000+	0	0.0	0.0
10,000-19,999 → Sub-7,500	2	0.3	2.7
10,000-19,999 → 7,500-9,999	5	0.7	6.8
10,000-19,999 → 20,000+	5	0.7	6.8
20,000+ → Sub-7,500	0	0.0	0.0
20,000+ → 7,500-9,999	0	0.0	0.0
20,000+ → 10,000-19,999	3	0.4	4.1

4.3.3 MAPE

MAPE is the primary accuracy benchmark used by both FHWA and MN DOT, making it the most direct basis for cross-study comparison. Values below roughly 10% are generally within acceptable bounds for medium-to-high-volume roads under FHWA Table 14 guidance, though the applicable threshold varies with sample size. Iowa estimates exceed MN DOT values in all non-missing buckets, with the gap widening at lower volumes.

Table 9: MAPE by AADT Bucket — 2022

AADT Bucket	n	MAPE (%)	MN MAPE (%)
Sub-7,500	394	47.47	NA
(7,500-9,999)	93	12.96	12.16
(10,000-19,999)	157	10.98	6.28
(20,000+)	59	8.59	5.81

Table 10: MAPE by AADT Bucket — 2024

AADT Bucket	n	MAPE (%)	MN MAPE (%)
Sub-7,500	400	38.71	NA
(7,500-9,999)	92	15.29	12.16
(10,000-19,999)	156	9.57	6.28
(20,000+)	60	6.79	5.81

4.3.4 Signed Percent Error (SPE)

Signed percent error captures the direction of bias — whether StreetLight tends to overestimate or underestimate relative to Iowa DOT counts. Positive values indicate overestimation. Iowa estimates are positive across all buckets and years, in contrast to MN DOT's mixed-sign pattern, suggesting a consistent directional tendency that becomes more pronounced at lower volumes.

Table 11: SPE by AADT Bucket — 2022

AADT Bucket	n	Signed % Error	MN Signed % Error
Sub-7,500	394	36.11	NA
(7,500-9,999)	93	7.34	-4.02
(10,000-19,999)	157	5.75	0.26
(20,000+)	59	4.63	-3.25

Table 12: SPE by AADT Bucket — 2024

AADT Bucket	n	Signed % Error	MN Signed % Error
Sub-7,500	400	32.12	NA
(7,500-9,999)	92	10.18	-4.02
(10,000-19,999)	156	4.79	0.26
(20,000+)	60	4.25	-3.25

4.3.5 Absolute Percent Error Standard Deviation (APE SD)

APE SD measures consistency across segments — a low value alongside low MAPE indicates reliable performance not just on average but across individual locations. Iowa APE SD values are substantially higher than MN DOT's in all reported buckets, indicating that even where average accuracy is acceptable, there is considerable segment-to-segment variability. The 2022 Sub-7,500 value of 150.3% is likely driven by one or more high-leverage outliers rather than a general pattern; see Section 5.2.

Table 13: APE SD by AADT Bucket — 2022

AADT Bucket	n	APE SD (%)	MN APE SD (%)
Sub-7,500	394	150.29	NA
(7,500-9,999)	93	18.46	8.30
(10,000-19,999)	157	13.76	4.98
(20,000+)	59	9.13	4.42

Table 14: APE SD by AADT Bucket — 2024

AADT Bucket	n	APE SD (%)	MN APE SD (%)
Sub-7,500	400	46.60	NA
(7,500-9,999)	92	21.96	8.30
(10,000-19,999)	156	11.94	4.98
(20,000+)	60	6.87	4.42

4.4 Additional Metrics

4.4.1 Median Percent Error (MPE)

MPE is complementary to the mean signed percent error by providing a more robust measure of typical bias — one less sensitive to segments with extreme over- or underestimates. Where the mean and median diverge substantially within a bucket, this signals that a small number of segments are disproportionately driving the average, which is particularly relevant in the Sub-7,500 bucket.

Table 15: MPE by AADT Bucket — 2022

AADT Bucket	n	Median % Error
Sub-7,500	394	13.45
(7,500-9,999)	93	4.21
(10,000-19,999)	157	6.87
(20,000+)	59	4.41

Table 16: MPE by AADT Bucket — 2024

AADT Bucket	n	Median % Error
Sub-7,500	400	23.99
(7,500-9,999)	92	8.01
(10,000-19,999)	156	4.95
(20,000+)	60	3.04

4.4.2 NRMSE

NRMSE provides a scale-normalized measure of overall error that penalizes large deviations more heavily than MAPE. Because it squares individual errors before averaging, it is more sensitive to outlier segments than MAPE is — so a notably higher NRMSE relative to MAPE within the same bucket is a signal that a small number of segments are contributing disproportionate error. The Sub-7,500 values (49–66%) illustrate this clearly.

Table 17: NRMSE by AADT Bucket — 2022

AADT Bucket	n	NRMSE (%)
Sub-7,500	394	66.25
(7,500-9,999)	93	22.76
(10,000-19,999)	157	16.31
(20,000+)	59	10.70

Table 18: NRMSE by AADT Bucket — 2024

AADT Bucket	n	NRMSE (%)
Sub-7,500	400	49.14
(7,500-9,999)	92	25.63
(10,000-19,999)	156	14.62
(20,000+)	60	8.96

4.5 Metrics

4.5.1 Mean Absolute Percentage Error (MAPE)

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{\text{Streetlight}_i - \text{AADT}_i}{\text{AADT}_i}\right| \times 100$$

MAPE measures average accuracy as a percentage of the reference (IADOT) count, ignoring the direction of error. Each segment's absolute percentage deviation from the ground-truth AADT is averaged across all segments in the bucket. Lower values indicate better accuracy; a MAPE of 0 would indicate perfect agreement. MAPE is used here for consistency with FHWA and MN DOT benchmarks, though it can become unstable at low AADT levels where small absolute differences translate to large percentage errors — a limitation most relevant to the Sub-7,500 bucket.

4.5.2 Mean Signed Percentage Error (Bias)

$$\text{Mean Signed \% Error} = \frac{1}{n}\sum_{i=1}^{n}\frac{\text{Streetlight}_i - \text{AADT}_i}{\text{AADT}_i} \times 100$$

Unlike MAPE, signed errors are not absolute — overestimates and underestimates offset one another. A positive mean indicates that StreetLight tends to *overestimate* volume relative to IADOT counts; a negative mean indicates systematic *underestimation*. Values near zero suggest that errors are roughly balanced in direction, even if individual segment-level errors are large.

4.5.3 Median Percentage Error

$$\text{Median \% Error} = \text{median}\left(\frac{\text{Streetlight}_i - \text{AADT}_i}{\text{AADT}_i} \times 100\right)$$

The median signed percentage error provides a more robust measure of central tendency than the mean, as it is less sensitive to extreme over- or underestimates on individual segments. It is interpreted in the same directional sense as the mean signed percentage error, but better reflects the *typical* segment's bias when the distribution of errors is skewed.

4.5.4 APE Standard Deviation

$$\text{APE SD} = \text{sd}\left(\left|\frac{\text{Streetlight}_i - \text{AADT}_i}{\text{AADT}_i}\right| \times 100\right)$$

The standard deviation of the absolute percentage errors measures the *consistency* of accuracy across segments within a bucket. A low APE SD alongside a low MAPE indicates that StreetLight performs reliably across segments; a high APE SD suggests that accuracy varies considerably from segment to segment, even if the average error is acceptable.

4.5.5 Normalized Root Mean Square Error (NRMSE)

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(\text{Streetlight}_i - \text{AADT}_i)^2}}{\overline{\text{AADT}}} \times 100$$

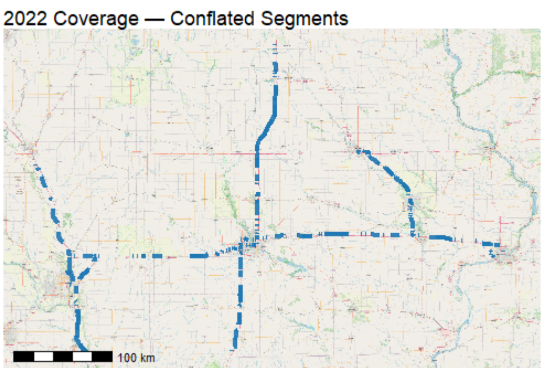
NRMSE penalizes large errors more heavily than MAPE by squaring deviations before averaging, making it sensitive to outlier segments with especially poor agreement. Normalizing by the mean AADT within the bucket expresses the error as a percentage of typical volume, enabling comparison across buckets with different volume scales.

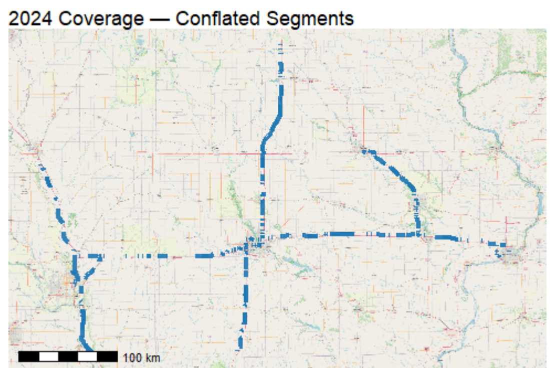
Note: *Equations taken from StreetLight AADT 2024 – Initial Release: Methodology and Validation White Paper*, page 5

5 Segment-Level Diagnostics

The analyses in this section move from aggregate bucket-level summaries to individual segment behavior. Two complementary diagnostics are presented: segments with persistently poor agreement (flagged as potential data quality or alignment issues) and segments with persistently strong agreement (natural candidates for an ongoing monitoring strategy). Together, these provide a concrete basis for evaluating where StreetLight estimates can be relied upon and where additional scrutiny is warranted.

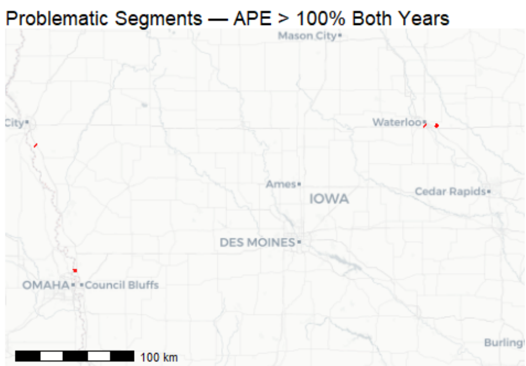
5.1 Initial Metrics Coverage





5.2 Beyond 100% - Bad Matches

Segments where StreetLight APE exceeds 100% in *both* 2022 and 2024 are flagged below. Requiring the threshold to be exceeded in both years reduces the chance that a segment is flagged due to a single-year data anomaly rather than a persistent discrepancy. A threshold of 100% indicates StreetLight volume is more than double (or less than half) the IADOT count; this conservative threshold is used here to surface only the most severe and persistent discrepancies. Segments flagged under this criterion likely reflect one or more of the following: low probe-vehicle penetration on a given corridor, complex traffic patterns near interchanges or ramps that challenge segment-level attribution, or misalignment between StreetLight and IADOT segment boundary definitions.



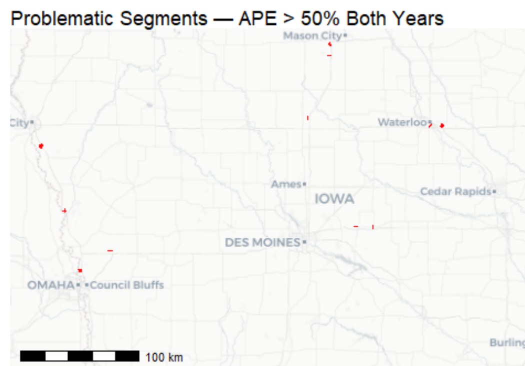


Table 19: Problematic Segments (APE > 100% in Both Years)

ID	Deviation - Overall	Deviation - 2022	Deviation - 2024
1,196,524,587	170.3	183.6	157.0
1,196,573,319	146.4	149.3	143.5
1,196,999,055	139.8	147.4	132.3
1,212,754,083	135.9	142.1	129.7
1,231,482,889	134.2	120.0	148.3
1,213,027,848	122.7	129.3	116.1
1,197,055,038	121.3	107.9	134.7

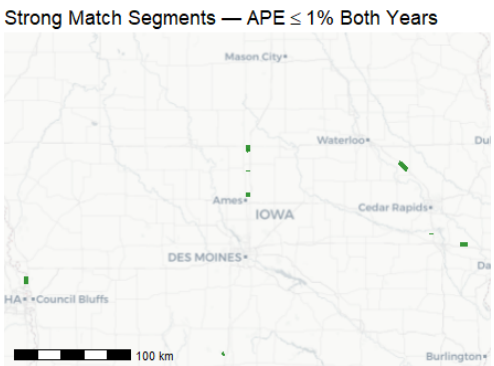
Table 20: Percent Deviations — Problematic Segments (APE > 100% in Both Years)

	2022		2024		
ID	APE - 2022	PE - 2022	APE - 2024	PE - 2024	Deviation - Overall
1,196,524,587	183.6	183.6	157.0	157.0	170.3
1,196,573,319	149.3	149.3	143.5	143.5	146.4
1,196,999,055	147.4	147.4	132.3	132.3	139.8
1,212,754,083	142.1	142.1	129.7	129.7	135.9
1,231,482,889	120.0	120.0	148.3	148.3	134.2
1,213,027,848	129.3	129.3	116.1	116.1	122.7
1,197,055,038	107.9	107.9	134.7	134.7	121.3

5.3 Within 5% - Strong Matches

Segments where StreetLight APE is within 5% in *both* 2022 and 2024 are highlighted below.

A threshold of 5% indicates that StreetLight volume is within a small margin of the IADOT count in both years, identifying segments where the estimates consistently match observed counts closely.



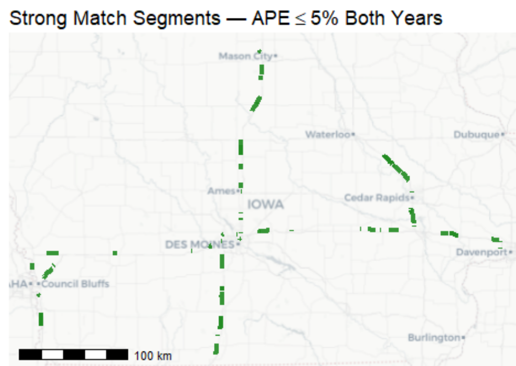


Table 21: Segment Match Counts by APE Threshold

APE Threshold	Matches in 2022	Matches in 2024	Matches in Both
<= 1%	37	30	10
<= 5%	164	166	92
<= 10%	304	292	206

Table 22: Expected vs. Observed AADT — Segments Matching in Both Years (APE <= 1%)

	2022		2024	
Segment	Expected	Observed	Expected	Observed
1,202,772,023	12,450	12,458	13,050	13,180
1,202,772,121	150	151	160	160
1,206,831,638	7,700	7,754	7,950	7,883
1,212,664,211	8,950	9,038	9,200	9,233
1,212,930,491	15,600	15,708	16,100	16,202
1,230,164,976	10,400	10,422	11,400	11,428
1,233,455,436	23,600	23,434	24,900	25,058
1,233,846,078	13,400	13,375	13,700	13,586
1,236,902,985	19,500	19,609	20,200	20,013
1,237,053,600	700	695	870	871

5.4 Recommended Monitoring Segments

The segment-level diagnostics point to a concrete path for preserving long-term validation capability. The core trade-off Iowa DOT faces — reducing tube-count costs versus maintaining an independent accuracy check — does not require an all-or-nothing decision. A small, well-chosen set of anchor segments can serve as an ongoing monitoring baseline, making it possible to detect future degradation in StreetLight performance without sustaining the full cost of current tube density. The goal is a small set of anchor segments that are geographically distributed, representative of the volume ranges of interest, and for which StreetLight accuracy is already well-established, providing some assurance that future deviations represent meaningful signals rather than noise.

We suggest that candidate segments should satisfy:

- APE $\leq$ 1% in both 2022 and 2024 (a demanding threshold that identifies segments with exceptionally consistent agreement across measurement cycles)
- AADT $\geq$ 7,500 (higher-volume corridors where StreetLight performs reliably and where counts matter most for planning purposes)
- Geographic spread across the coverage area (not clustered on a single corridor)
- Proximity to existing tube infrastructure (minimizes setup burden and safety risk on high-speed segments)

Of the 10 segments in Table 22 meeting the $\leq$ 1% threshold in both years, 8 satisfy the AADT $\geq$ 7,500 criterion; the remaining two (segments 1,202,772,121 and 1,237,053,600) fall well below this threshold and should be excluded from consideration as monitoring candidates. These 8 segments form a natural, conservative starting pool. A final selection of 3–5 segments from this (or an alternatively specified) pool — chosen to maximize geographic spread and volume-range coverage — would provide an ongoing validation baseline at substantially reduced cost relative to current tube density. Analysts preferring a broader candidate pool may relax the APE threshold to $\leq$ 5%, which remains within FHWA acceptable bounds; however, this expands the pool to roughly 92 segments and requires additional filtering on geographic and volume criteria to arrive at a manageable monitoring set.

This approach does not eliminate uncertainty about StreetLight's use of Iowa DOT data during calibration. However, it preserves the more important operational capability: the ability to detect future degradation in StreetLight accuracy using an independent validation baseline. Final segment selection from this pool is left to Iowa DOT, based on operational knowledge of existing tube placement, access constraints, and corridor priorities not captured in the data alone.